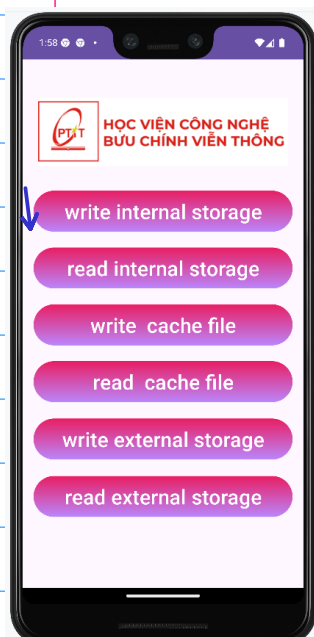


```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android">

    <corners android:radius="50dp"/>
    <gradient android:startColor="#E91E63"
        android:endColor="@color/purple_200"
        android:angle="200"
    />
</shape>
```

`android:background="@drawable/buttonround"`

`app:backgroundTint="@null"`

```
public class MainActivity extends AppCompatActivity implements View.OnClickListener
```

```
    findViewById(R.id.btnwriteIS).setOnClickListener(this);  
    findViewById(R.id.btnreadIS).setOnClickListener(this);  
    findViewById(R.id.btnwriteFC).setOnClickListener(this);  
    findViewById(R.id.btnreadFC).setOnClickListener(this);  
    findViewById(R.id.btnwriteES).setOnClickListener(this);  
    findViewById(R.id.btnreadES).setOnClickListener(this);
```

```
@Override  
public void onClick(View view) {  
    switch (view.getId()) {  
        case R.id.btnreadIS:  
            String readDataIS = readIS( filename: "myfile.txt");  
            Toast.makeText( context: this, text: "Nội dung file:"  
                + readDataIS, Toast.LENGTH_SHORT).show();  
            break;  
        case R.id.btnwriteIS:  
            writeIS( filename: "myfile.txt", data: "Xin chào các bạn!");  
            break;  
        case R.id.btnreadFC:  
            String readDataFC = readFC( filename: "mycached.cache");  
            Toast.makeText( context: this, text: "Data cache: " +  
                readDataFC, Toast.LENGTH_SHORT).show();  
            break;  
        case R.id.btnwriteFC:  
            writeFC( filename: "mycached.cache", data: "Dữ liệu lưu vào cahce");  
            break;  
        case R.id.btnwriteES:  
            writeES( filename: "mysdcard.txt", data: "Dữ liệu lưu vào thẻ nhớ ngoài");  
            break;  
        case R.id.btnreadES:  
            String readDataEX = readES( filename: "mysdcard.txt");  
            Toast.makeText( context: this, text: "Data sdcard: " +  
                readDataEX, Toast.LENGTH_SHORT).show();  
            break;  
    }  
}
```

```
        case R.id.btnwriteIS:  
            writeIS( filename: "myfile.txt", data: "Xin chào các bạn!");  
            break;
```

```
private void writeIS(String filename,String data) { 1 usage  
  
    try {  
  
        FileOutputStream fout = openFileOutput(filename,MODE_PRIVATE);  
        fout.write(data.getBytes(StandardCharsets.UTF_8));  
        fout.close();  
        Toast.makeText( context: this, text: "Dữ liệu đã được ghi", Toast.LENGTH_SHORT).show();  
    } catch ( Exception ex )  
    {  
        Toast.makeText( context: this, text: "Error:" + ex.getMessage(), Toast.LENGTH_SHORT).show();  
        ex.printStackTrace();  
    }  
}
```

```

case R.id.btnreadIS:
    String readDataIS = readIS( filename: "myfile.txt");
    Toast.makeText( context: this, text: "Nội dung file:"
        + readDataIS, Toast.LENGTH_SHORT).show();
    break;

```

```

private String readIS(String filename) { 1usage
    try (FileInputStream fin = openFileInput(filename))
    {
        byte[] buffer = new byte[1024];
        int length = 0;
        length = fin.read(buffer);
        return new String(buffer, offset: 0, length);
    } catch ( Exception ex ) {
        ex.printStackTrace();
    }
    return null;
}

```

```

case R.id.btnreadFC:
    String readDataFC = readFC( filename: "mycached.cache");
    Toast.makeText( context: this, text: "Data cache: " +
        readDataFC, Toast.LENGTH_SHORT).show();
    break;

```

```

private void writeFC(String filename, String data) { 1usage
    File cacheDir = getCacheDir();
    try (FileOutputStream fout = new FileOutputStream(new File(cacheDir,filename))) {
        fout.write(data.getBytes(StandardCharsets.UTF_8));
        Toast.makeText( context: this, text: "Dữ liệu lưu vào cache thành công!",
            Toast.LENGTH_SHORT).show();
    } catch ( Exception ex )
    {
        ex.printStackTrace();
        Toast.makeText( context: this, text: "Error:" + ex.getMessage(), Toast.LENGTH_SHORT).show();
    }
}

```

```

private String readFC(String filename) { 1usage
    File cacheDir = getCacheDir();
    byte[] buffer = new byte[1024];
    int length = 0;
    try (FileInputStream fin= new FileInputStream(new File(cacheDir, filename))) {
        length = fin.read(buffer);
        return new String(buffer, offset: 0, length);
    } catch ( Exception ex ) {
        ex.printStackTrace();
        Toast.makeText( context: this, text: "Error: " +
            ex.getMessage(), Toast.LENGTH_SHORT).show();
    }
    return null;
}

```

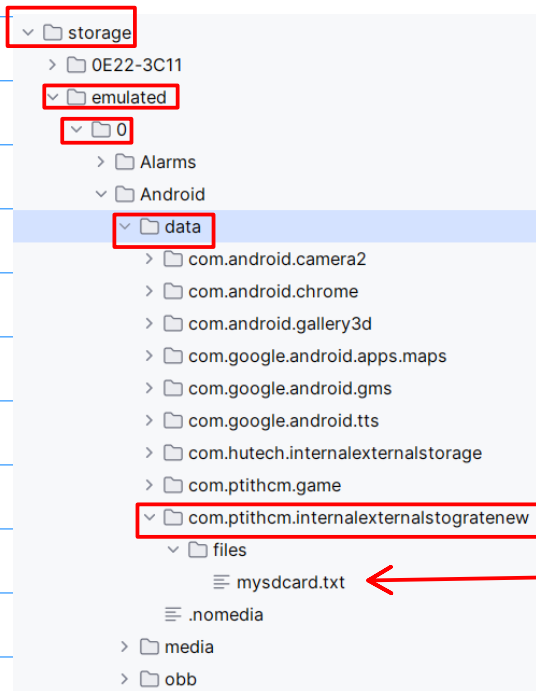
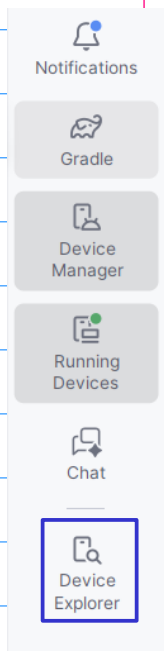
```
private void writeES(String filename, String data) { 1 usage

String sdcard = getExternalFilesDir( type: "" ) + "/" + filename;

try {
    OutputStreamWriter writer = new OutputStreamWriter(new FileOutputStream(sdcard));
    writer.write(data);
    writer.close();
    Toast.makeText( context: this, text: "Lưu thành công", Toast.LENGTH_SHORT).show();
} catch ( Exception ex )
{
    ex.printStackTrace();
}
```

```
private String readES(String filename) { 1 usage

String sdcard = getExternalFilesDir( type: "" ) + "/" + filename;
try {
    Scanner scan = new Scanner(new File(sdcard));
    String data="";
    while (scan.hasNext()){
        data = data + scan.nextLine() + "\n";
    }
    scan.close();
    return data;
} catch ( Exception ex )
{
    ex.printStackTrace();
}
```



primitive => int, float, double

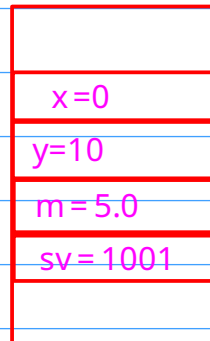
non-primitive -> Integer, Float, SinhVien

```
SinhVien{  
    int x,y=10;  
    double m=5;  
    private String MSSV;  
    private String Hoten;  
    Private int Diem  
}
```

```
SinhVien sv, sv1;  
sv= new SinhVien();
```

Java Stack

sv1=sv;



1001

SinhVien

MSSV=""

Hoten=""

Diem=0

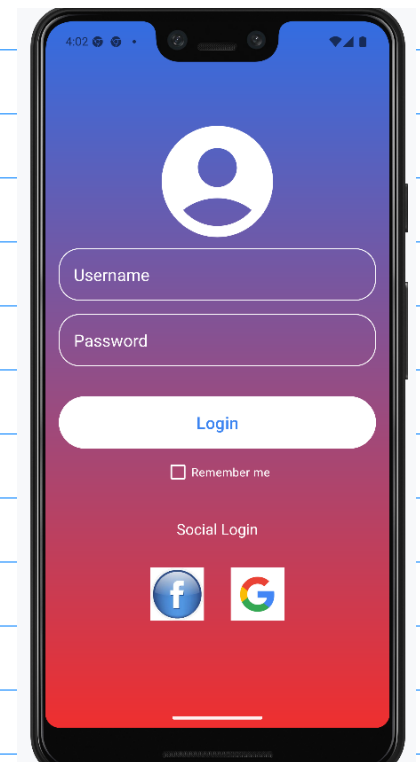
LS LoginShareReference

</> login_background.xml

```
<?xml version="1.0" encoding="utf-8"?>  
<shape xmlns:android="http://schemas.android.com/apk/res/android">  
    <gradient android:type="linear" android:angle="100"  
        android:startColor="#336De1"  
        android:endColor="#EF2f2f" />  
</shape>
```

</> round_conner.xml

```
<?xml version="1.0" encoding="utf-8"?>  
<shape xmlns:android="http://schemas.android.com/apk/res/android"  
    android:shape="rectangle">  
    <corners android:radius="24dp" />  
    <stroke android:width="1dp" android:color="@color/white" />  
</shape>
```



```
SharedPreferences sharedPreferences = getSharedPreferences( name: "login_check",MODE_PRIVATE);
SharedPreferences.Editor editor = sharedPreferences.edit();
editor.putString( S: "login", s1: "true");
editor.apply();
```

```
Intent intent = new Intent( packageContext: MainActivity.this,MainActivity2.class);
startActivity(intent);
```

```
private void checkLogin() 1 usage
{
    SharedPreferences sharedPreferences = getSharedPreferences( name: "login_check",MODE_PRIVATE);
    String login = sharedPreferences.getString( S: "login", s1: "");
    if(login.equals("true"))
    {
        Intent intent = new Intent( packageContext: MainActivity.this,MainActivity2.class);
        startActivity(intent);
    }
}
```